

Student Performance Analysis Using SQL and Python

A Data Analysis Project

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Project Overview & Objectives



Why This Matters: Understanding factors associated with student performance enables early identification of at-risk students and targeted academic interventions.

- Understand dataset structure and quality
- Analyse final grades and their distribution
- Examine factors associated with academic performance
- Compare student groups by gender and school
- Identify academically at-risk students
- Demonstrate SQL + Python analytics workflow

Dataset Overview

395

Student Records

Complete valid records with no missing values

33

Variables

Academic, behavioral, and demographic factors

10.42

Avg Final Grade

Mean G3 score across all 395 students

0

Missing Values

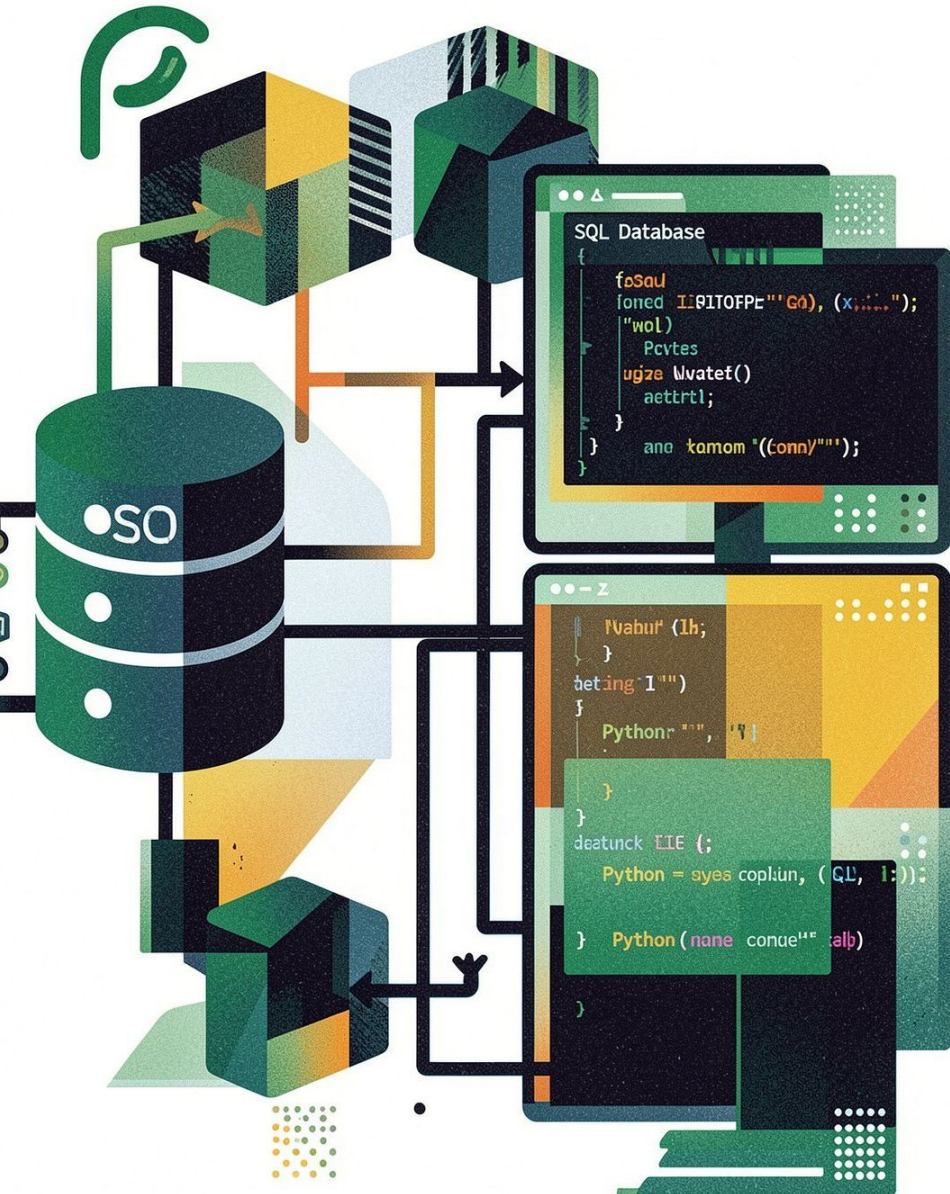
Dataset ready for analysis immediately

Primary Outcome Variable

G3 – Final Grade is the main outcome of interest, supported by G1 (Period 1) and G2 (Period 2) as predictive indicators.

Key Variables Analysed

- G1, G2, G3 – Period grades
- Study Time, Previous Failures, Absences
- Gender and School (GP vs MS)



Data Analytics Workflow

Dataset

Data Loading

Quality Check

SQL Exploration

MySQL

Structured queries,
aggregation, grouping

Python / Pandas

Data loading, cleaning,
statistical analysis

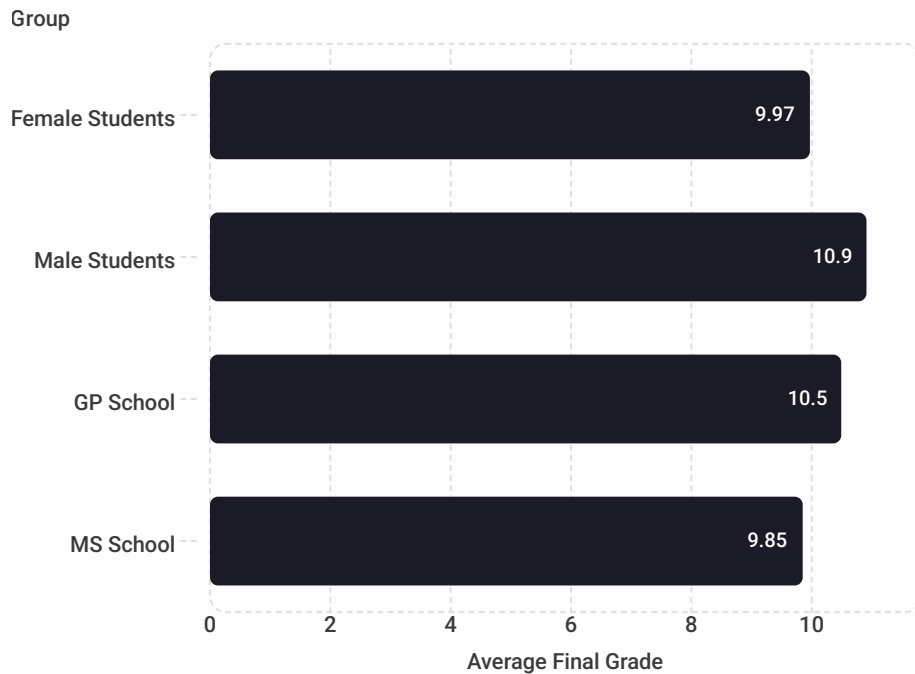
Matplotlib

Data visualization and
charts

Google Colab

Cloud-based development environment

Group Comparisons: Gender & School



Gender Comparison

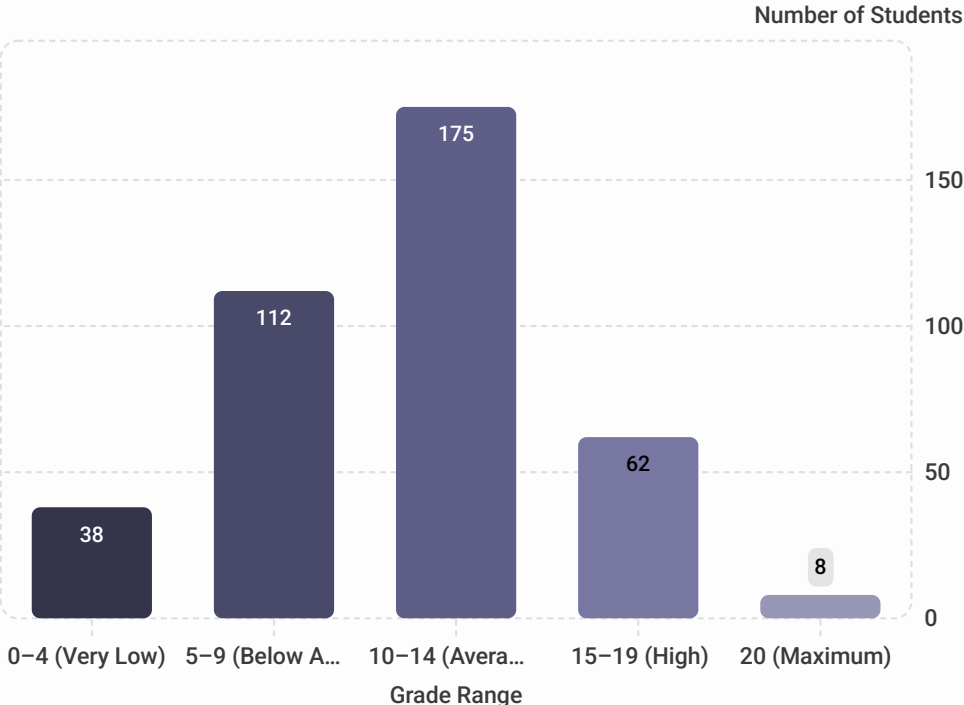
- **Female Students:** Average Final Grade ≈ 9.97
- **Male Students:** Average Final Grade ≈ 10.91

School Comparison

- **GP School:** Average Final Grade ≈ 10.49
- **MS School:** Average Final Grade ≈ 9.85

📄 Observed differences describe this dataset only and do not establish causation.

Overall Final Grade Distribution



Key Descriptive Statistics

Most students cluster around the middle of the grading scale, with smaller groups at the extremes – both very high and very low performers.

Average Final Grade

10.42 – midpoint of the 0–20 scale

Highest Grade

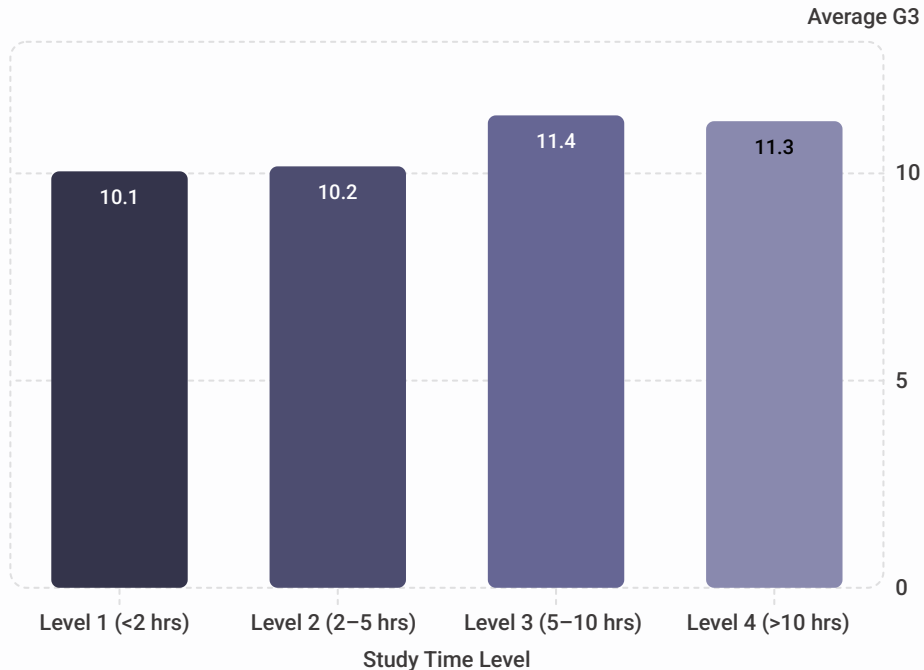
20 – maximum possible score achieved

Lowest Grade

0 – minimum observed in dataset

Study Time & Final Grade

Average G3 by Study Time Level



What the Data Shows

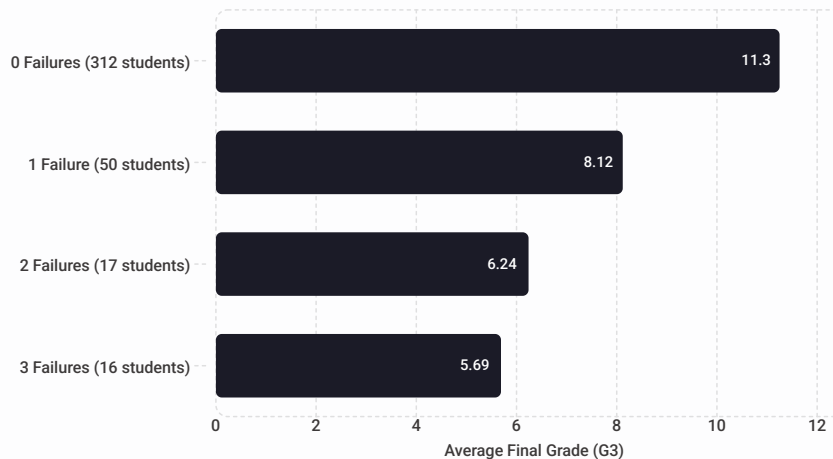
- Higher study time levels associate with modestly better average grades at group level
- Level 3 (5–10 hrs) yields the highest average G3 at **11.40**
- Level 4 shows a slight dip to **11.26** – not a strict monotonic relationship

📄 **Correlation with G3: 0.0978** – Study time shows some group-level association but is not a strong predictor by itself.



Previous Failures: A Strong Risk Signal

Previous Failures



Correlation with G3

-0.3604

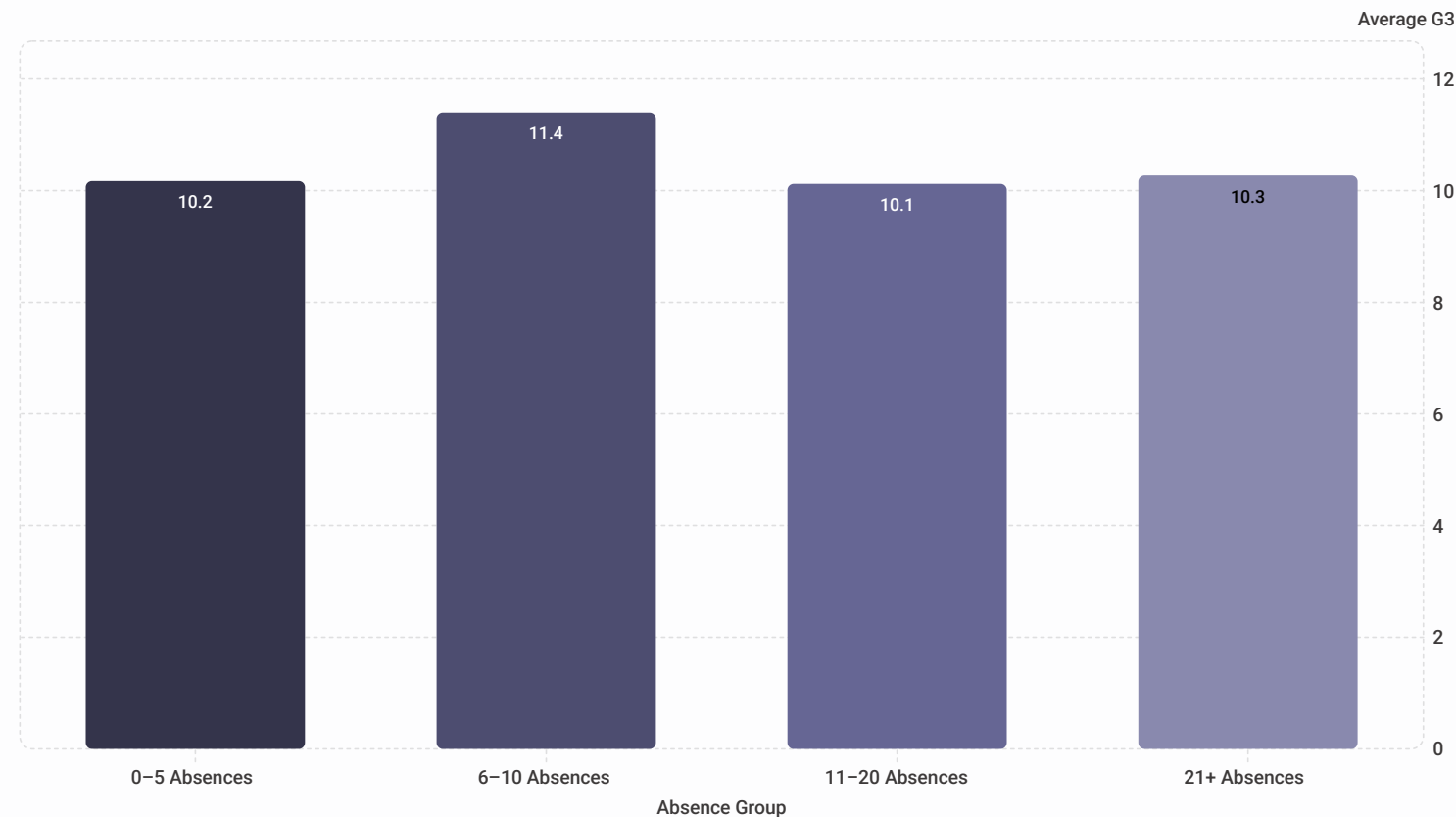
Strong negative relationship — the strongest risk indicator identified in this project.

Critical Insight

Students with prior failures score dramatically lower on average. A student with 3 previous failures averages just **5.69** — less than half the score of students with no prior failures (**11.25**). Previous failure history is the single most actionable early warning signal in this dataset.

Absences & Previous Grades Analysis

Absences by Group — Average G3



Absence correlation with G3: **0.0342** — extremely weak, no clear linear relationship observed.

Correlation with Final Grade (G3)



G2 — Period 2 Grade
Correlation: **0.9049** ★ Strongest predictor



Study Time
Correlation: **0.0978** — weak



G1 — Period 1 Grade
Correlation: **0.8015** ★ Strong predictor

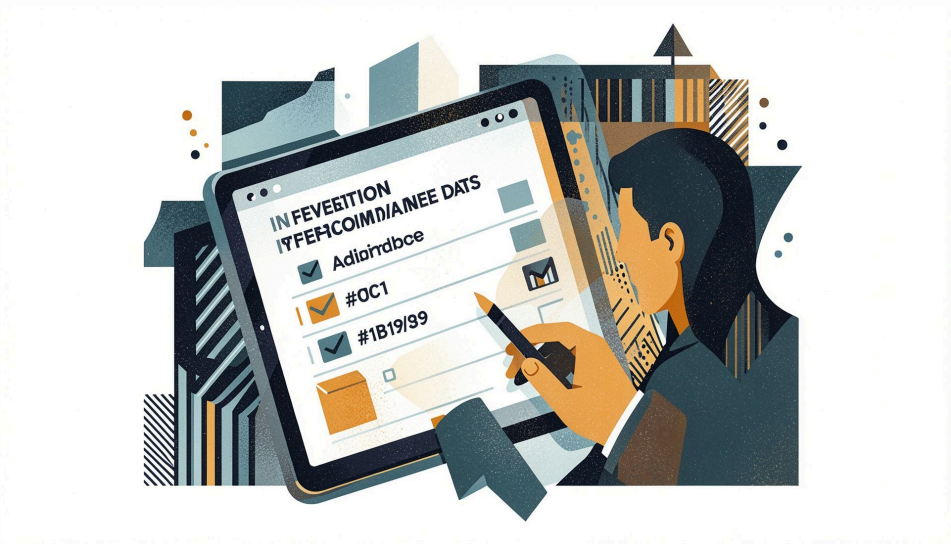


Absences
Correlation: **0.0342** — extremely weak



Previous Failures
Correlation: **-0.3604** (negative)

At-Risk Classification & Key Findings



At-Risk Student Classification

Students scoring below a defined threshold were classified as academically at-risk, enabling targeted early intervention.

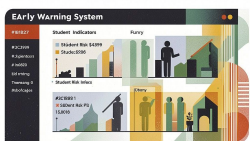
- ❑ Correlation ≠ causation. All findings describe observed associations within this dataset and should not be interpreted as causal mechanisms.

32.91%
Proportion of students classified as at-risk

G2 → G3
Correlation 0.9049 — most reliable predictor

Failures
Correlation -0.3604 — strongest risk signal

Conclusions & Recommendations



Monitor Early Grades

G1 and G2 are the strongest predictors of final outcomes ($r = 0.80$ and 0.90). Early grade monitoring enables timely intervention before G3 is finalised.



Prioritise Failure History

Previous failures carry the strongest risk signal ($r = -0.3604$). Students with even one prior failure should receive structured academic support immediately.



SQL + Python Workflow Value

The combined SQL exploration and Python statistical analysis workflow proved effective for identifying actionable patterns across 395 student records and 33 variables.

- ❏ All findings are descriptive and associational. Future work could explore inferential statistics or predictive modelling to further validate these patterns.